Pre Analysis Plan - RT

Given we have access to a subset of the data, we rely on the Stata do files as our main pre-commitment to specific analysis.

We now describe your general data cleaning approach:

1. Set all negative values to 'missing', as the SOEP uses negative values to explain the reason for why a response is not recorded. By setting them to missing, Stata will not include those observations in the statistical tests.
2. We create a variable that is 0 if a particular individual (by pid) never reports having any child in any wave of the survey they are observed in, and 1 otherwise. We drop all individuals that have never had a child (i.e., where this variable is 0).
3. We create a variable that is 1 for an individual in a given year if they report ever having a daughter. That is, for an individual, this variable can be both 0 or 1 depending on the year or when/if they have their first daughter. This variable is our chief variable of interest for all hypotheses except for H4 (where it is the interaction of this variable and if the respondent is male that is of interest).

For our identification strategy in general, we assume the gender of a child is essentially randomly assigned. The choice to actually have a child at some stage is essentially endogenous, so to control for this we drop all individuals who are never observed to have a child. How many children is also endogenous, but for power considerations we include the number of children as an independent variable (with dummy variables for each specific number of children), instead of focusing on one specific number of children. We additionally include other independent variables for controls, such as the respondent's age (and an age-squared term for non-linear effects), the respondent's gender, the year of the survey (dummy variable for each year), whether the respondent was born in Germany or not, and whether the respondent currently lives in East Germany or not. All of these independent variables could plausibly affect our dependent variable, and the related questions are included in most waves. All of our hypotheses include these variables as controls (even if they are not present, e.g. they might be missing in specific waves).

We now describe any specific data cleaning / generating processes for the individual hypotheses (if any), as well as their statistical tests.

H1

Having daughters leads to stronger support for gender equality (as measured by attitudes)

Our dependent variable is 'plj0625: position of women: husband should have the final say'. We selected this variable as we find it has slightly more valid entries than the other variables in the set we consider appropriate measures for attitudes towards gender equality. We were disappointed by the number of waves the relevant questions were asked in, and we consider other possible measures as not-sufficiently close to the concept of *attitudes* towards gender equality (unlike behaviors or outcomes in H3). However, we did not select the hypotheses we are to test – given the provided dataset we believe this to be the most appropriate variable.

As the variable is on a Likert scale, for the statistical test we use an ordered logit (see Liddell & Kruschke 2018 https://doi.org/10.1016/j.jesp.2018.08.009). We do not use the panel version of this test (xtologit) as the dependent variable was only asked in one wave. The Stata command is as follows:

ologit plj0625 daughter i.sumkids age age2 i.sex i.syear i.germborn i.sampreg, vce(cluster pid)

The coefficient on the *daughter* variable is the relevant one for this hypothesis. As it is on how much they agree with the husband having the final say, we interpret lower values of this variable as stronger attitudes towards gender equality. Therefore, this hypothesis predicts a negative coefficient on the *daughter* variable.

H2

Having daughters leads to more left-leaning political preferences

An immediately obvious variable to test this hypothesis would be 'plh0004: Political attitude', however, there are other possible variables that have been asked in more waves of the SOEP. We use 'plj0046: Worried About Immigration To Germany', and assume people that are less worried about immigration are more left leaning. We use a panel data ordered logit, as the variable is again on a Likert scale. The Stata command is as follows:

xtset pid syear

xtologit plj0046 daughter i.sumkids age age2 i.sex i.syear i.germborn i.sampreg, vce(cluster pid)

As the responses are something like 1: very concerned, 2: somewhat concerned, and 3: not concerned at all, then we consider a higher value of this variable is associated with left-leaning political preferences. The relevant coefficient is again that which is associated with the *daughter* variable. A positive coefficient on this variable would be in line with this hypothesis.

H3

Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

For this hypothesis, we consider the variable 'pli0043_h : Hours Weekday Housework', which is the number of hours per day the respondent reports spending on housework. Historically, this would have been a gendered task, so more equal amounts of time spent on doing housework would be associated with more gender equality.

We perform additional data cleaning and exclude more observations for this particular hypothesis. In particular, we focus on couples that live in the same household (i.e. their housework is a public good) and that are of mixed-gender (as this is about gender equality). We pair couples based on parid, and code if they live in the same household if their hid is the same or not. We drop observations at the year level if they do not live in the same household as a partner (they could be single or they could live in a different household than their partner). We also drop observations at the year level if the couple is of the same gender.

We then turn our focus to generating the dependent variable based on pli0043_h. Given this measure is hours spent per day on housework, we can see there are some unreasonable responses like 24 hours. We drop all observations where pli0043_h is greater than 8 hours, which is approximately the 99th percentile, to deal with these extreme outliers. We then generate a ratio of housework at the individual level by calculating pli0043_own/(pli0043_own+pli0043_partner), and as we are interested in more equal allocations of housework, we take the absolute value of the difference from 0.5. Missing values propagate in Stata, in that if pli0043 is missing for one of the couple, then the ratio will be missing as well. Finally, to avoid double counting a couple (e.g. if one person has .4 for the ratio, then their partner would have .6), we drop all men from our dataset. We then conduct the following panel regression:

xtset pid syear

xtreg pli0043_abs daughter i.sumkids age age2 i.syear i.germborn i.sampreg, vce(cluster pid)

More gender equality would imply *pli0043_abs* is smaller, as this variable is the absolute difference of the ratio of housework and an equal split. The relevant coefficient is again that which is associated with the *daughter* variable. A negative coefficient on this variable would be in line with this hypothesis.

H4

Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

We follow the same procedure as H1 but include an interaction term of the daughter variable with a dummy variable for if the respondent is a male. That is, we run the following in Stata:

ologit plj0625 daughter daughterXmale i.sumkids age age2 i.sex i.syear i.germborn i.sampreg, vce(cluster pid)

Unlike the other hypotheses, it is now the daughterXmale variable that is of interest. If the change in attitudes towards gender equality is greater for men, the associated coefficient should be negative.